SIMATS ENGINEERING

Department of Computer Science & Engineering

ASSESSMENT TOOL 1- INDUSTRY PROBLEM SOLVING TASK

Course Code:	CSA1223	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL 1- INDUSTRY PROBLEM SOLVING TASK
Weightage:	30%	Total Marks:	25
CO4 Statement:	*Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.*		
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Department:	B TECH AI/ML	Date:	24/08/26

83%

ANNEXURE A

INDUSTRY PROBLEM SOLVING TASK

1. **Smartphone Performance Optimization**

   A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.

2. **Cloud Server Memory Bottleneck**

   A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.

3. **Autonomous Vehicle Data Processing**

   An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.

4. **AI Inference Edge Device**

   An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.

5. High-Performance Gaming Console

A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

Code of Conduct:

I V.Adarsh Reddy *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

V.Adarsh Reddy

Signature of the student:

MARKS ALLOCATION

	Understanding of Industry Problem (20%)	Application of Engineering Concepts (25%)	Solution Design (25%)	Use of Tools (15%)	Analysis (10%)
Q1	1	1.25	1	0.75	0.5
Q2	1	1.25	1	0.5	0.5
Q3	0.5	1	1	0.5	0.5
Q4	0.5	1.25	1.25	0.75	0.5
Q5	1	1	1.25	0.5	0.5

20.75

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints(1)	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem(1.25)	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts